

```

1 # Create some doctors
2 doctor1 = Doctor(1, "MD", 10, "New York", "Cardiologist", num_days=14)
3 doctor2 = Doctor(2, "PhD", 15, "New York", "Cardiologist", num_days=5)
4 doctor3 = Doctor(3, "MD", 8, "Los Angeles", "Neurologist", num_days=18)
5 matching_system.add_doctor(doctor1)
6 matching_system.add_doctor(doctor2)
7 matching_system.add_doctor(doctor3)

```

[/usr/local/lib/python3.10/dist-packages/huggingface_hub/utils/_token.py:89](#): UserWarning:
The secret 'HF_TOKEN' does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/token>).
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
```

```

1_Pooling/config.json: 0%|          | 0.00/190 [00:00<?, ?B/s]

config.json: 0%|          | 0.00/612 [00:00<?, ?B/s]

model.safetensors: 0%|          | 0.00/90.9M [00:00<?, ?B/s]

tokenizer_config.json: 0%|          | 0.00/350 [00:00<?, ?B/s]

```

```

1 # Create some patients
2 patient1 = Patient(1, "New York", "Stable", "video1.mp4", "Heart palpitations and shortness of breath", "Cardiologist", 5)
3 patient2 = Patient(2, "Los Angeles", "Critical", "video2.mp4", "Severe headache and confusion", "Neurologist", 7, is_emergency=True)
4 patient3 = Patient(3, "New York", "Stable", "video3.mp4", "Chest pain and dizziness", "Cardiologist", 6)

```

[+ Code](#) [+ Markdown](#)

```

1 # Add patients to waiting list
2 appointment_system.add_to_waiting_list(patient1)
3 appointment_system.add_to_waiting_list(patient2)
4 appointment_system.add_to_waiting_list(patient3)

```

```

1 # Process appointments
2 appointment_system.process_appointments(matching_system)
3
4 # Update waiting times and reprocess
5 appointment_system.update_waiting_times()
6 appointment_system.process_appointments(matching_system)
7
8 # Simulate an emergency patient arrival
9 emergency_patient = Patient(4, "New York", "Critical", "video4.mp4", "Severe chest pain and shortness of breath", "Ca

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1 # Process appointments
2 appointment_system.process_appointments(matching_system)
3
4 # Update waiting times and reprocess
5 appointment_system.update_waiting_times()
6 appointment_system.process_appointments(matching_system)
7
8 # Simulate an emergency patient arrival
9 emergency_patient = Patient(4, "New York", "Critical", "video4.mp4", "Severe chest pain and shortness of breath", "Cardiologist", 7, is_emergency=True)
10 appointment_system.add_to_waiting_list(emergency_patient)
11 appointment_system.process_appointments(matching_system)

```

```

Scheduled appointment for patient 2 with doctor 3 at 2024-08-05 09:00:00
Waiting time: 0 days
Scheduled appointment for patient 3 with doctor 1 at 2024-08-05 09:00:00
Waiting time: 0 days
Scheduled appointment for patient 1 with doctor 1 at 2024-08-05 10:00:00
Waiting time: 0 days
Scheduled appointment for patient 4 with doctor 1 at 2024-08-05 11:00:00
Waiting time: 0 days

```

```
1 print(patient1.appointment)
```

```

1 # Generate 100 dummy patients
2 dummy_patients = generate_dummy_patients(1000)
3
4 # Print a few examples to verify
5 for patient in dummy_patients[:10]:
6     print(f"Patient {patient.id}: {patient.location}, {patient.health_status}, {patient.required_doctor_type}, "
7           f"Emergency: {patient.is_emergency}, Score: {patient.score}, Symptoms: {patient.text_description}")
8
9 print("Length of the dummy patients: ", len(dummy_patients))

```

Patient 1: San Antonio, Serious, Orthopedist, Emergency: False, Score: 7.0, Symptoms: Back pain and Osteoporosis
 Patient 2: Los Angeles, Critical, Pediatrician, Emergency: True, Score: 2.0, Symptoms: Sore throat and Rash
 Patient 3: Philadelphia, Stable, Gynecologist, Emergency: False, Score: 7.0, Symptoms: Vaginal discharge and Breast lumps
 Patient 4: San Diego, Critical, Psychiatrist, Emergency: True, Score: 7.0, Symptoms: Insomnia and Mood swings
 Patient 5: Houston, Critical, Psychiatrist, Emergency: True, Score: 6.0, Symptoms: Anxiety and Insomnia
 Patient 6: San Antonio, Critical, Endocrinologist, Emergency: True, Score: 4.0, Symptoms: Weight management issues and Hormonal imbalances
 Patient 7: New York, Stable, Urologist, Emergency: False, Score: 6.0, Symptoms: Erectile dysfunction and Blood in urine
 Patient 8: San Diego, Stable, Urologist, Emergency: False, Score: 3.0, Symptoms: Blood in urine and Erectile dysfunction
 Patient 9: San Diego, Moderate, Pediatrician, Emergency: False, Score: 2.0, Symptoms: Fever and Sore throat
 Patient 10: Houston, Critical, Gynecologist, Emergency: True, Score: 5.0, Symptoms: Breast lumps and Abnormal bleeding
 Length of the dummy patients: 1000

```

1 # Generate dummy doctors
2 dummy_doctors = generate_dummy_doctors(num_doctors=12)
3
4 # Assuming the matching_system and its add_doctor method are already defined
5 for doctor in dummy_doctors:
6     matching_system.add_doctor(doctor)
7
8 # Print a few examples to verify
9 for doctor in dummy_doctors:
10     print(f"Doctor {doctor.id}: {doctor.degree}, {doctor.experience} years, {doctor.location}, {doctor.specialization}")
11

```

Doctor 1: DO, 30 years, Chicago, Cardiologist
 Doctor 2: DO, 11 years, Philadelphia, Neurologist
 Doctor 3: PhD, 13 years, Philadelphia, Oncologist
 Doctor 4: PhD, 18 years, Houston, Pediatrician
 Doctor 5: MBBS, 19 years, Houston, Dermatologist
 Doctor 6: MBBS, 14 years, New York, Orthopedist
 Doctor 7: DO, 19 years, New York, Gynecologist
 Doctor 8: DO, 7 years, Los Angeles, Urologist
 Doctor 9: DO, 24 years, Houston, Psychiatrist
 Doctor 10: MD, 8 years, Chicago, Endocrinologist
 Doctor 11: PhD, 14 years, Los Angeles, Orthopedist
 Doctor 12: MD, 13 years, Chicago, Pediatrician

Scheduled appointment for patient 911 with doctor 10 at 2024-08-23 09:00:00
Waiting time: 17 days
Scheduled appointment for patient 115 with doctor 11 at 2024-08-16 10:00:00
Waiting time: 10 days
Scheduled appointment for patient 411 with doctor 2 at 2024-08-20 10:00:00
Waiting time: 14 days
Scheduled appointment for patient 959 with doctor 3 at 2024-08-06 12:00:00
Waiting time: 0 days
Scheduled appointment for patient 198 with doctor 11 at 2024-08-16 11:00:00
Waiting time: 10 days
Scheduled appointment for patient 963 with doctor 11 at 2024-08-16 12:00:00
Waiting time: 10 days
Scheduled appointment for patient 636 with doctor 1 at 2024-08-09 14:00:00
Waiting time: 3 days
Scheduled appointment for patient 668 with doctor 9 at 2024-08-21 15:00:00
Waiting time: 15 days
Scheduled appointment for patient 988 with doctor 3 at 2024-08-22 16:00:00
Waiting time: 17 days
Scheduled appointment for patient 125 with doctor 4 at 2024-08-09 09:00:00
Waiting time: 3 days
Scheduled appointment for patient 926 with doctor 1 at 2024-08-09 15:00:00
Waiting time: 3 days
Scheduled appointment for patient 202 with doctor 3 at 2024-08-23 09:00:00
Waiting time: 17 days
Scheduled appointment for patient 463 with doctor 3 at 2024-08-23 10:00:00
Waiting time: 17 days
Scheduled appointment for patient 228 with doctor 12 at 2024-08-15 13:00:00
Waiting time: 9 days
Scheduled appointment for patient 802 with doctor 3 at 2024-08-23 11:00:00
Waiting time: 17 days

```
1 # Checng Availability of the doctor
2 doctor1.view_schedule()
```

```
[(datetime.datetime(2024, 8, 5, 10, 0), 'Booked - Patient 475'),
 (datetime.datetime(2024, 8, 5, 11, 0), 'Booked - Patient 4'),
 (datetime.datetime(2024, 8, 5, 12, 0), 'Booked - Patient 817'),
 (datetime.datetime(2024, 8, 5, 13, 0), 'Booked - Patient 11'),
 (datetime.datetime(2024, 8, 5, 14, 0), 'Booked - Patient 735'),
 (datetime.datetime(2024, 8, 5, 15, 0), 'Booked - Patient 170'),
 (datetime.datetime(2024, 8, 5, 16, 0), 'Booked - Patient 331'),
 (datetime.datetime(2024, 8, 12, 9, 0), 'Booked - Patient 188'),
 (datetime.datetime(2024, 8, 12, 10, 0), 'Booked - Patient 613'),
 (datetime.datetime(2024, 8, 12, 11, 0), 'Booked - Patient 342'),
 (datetime.datetime(2024, 8, 12, 12, 0), 'Booked - Patient 607'),
 (datetime.datetime(2024, 8, 12, 13, 0), 'Booked - Patient 537'),
 (datetime.datetime(2024, 8, 12, 14, 0), 'Booked - Patient 943'),
 (datetime.datetime(2024, 8, 12, 15, 0), 'Booked - Patient 929'),
 (datetime.datetime(2024, 8, 12, 16, 0), 'Booked - Patient 395'),
 (datetime.datetime(2024, 8, 13, 9, 0), 'Booked - Patient 401'),
 (datetime.datetime(2024, 8, 13, 10, 0), 'Booked - Patient 761'),
 (datetime.datetime(2024, 8, 13, 11, 0), 'Booked - Patient 89'),
 (datetime.datetime(2024, 8, 13, 12, 0), 'Booked - Patient 132'),
 (datetime.datetime(2024, 8, 13, 13, 0), 'Booked - Patient 832'),
 (datetime.datetime(2024, 8, 13, 14, 0), 'Booked - Patient 322'),
 (datetime.datetime(2024, 8, 13, 15, 0), 'Booked - Patient 85'),
 (datetime.datetime(2024, 8, 13, 16, 0), 'Booked - Patient 21'),
 (datetime.datetime(2024, 8, 14, 9, 0), 'Booked - Patient 281'),
 (datetime.datetime(2024, 8, 14, 10, 0), 'Booked - Patient 82'),
 (datetime.datetime(2024, 8, 14, 11, 0), 'Booked - Patient 685'),
 (datetime.datetime(2024, 8, 14, 12, 0), 'Booked - Patient 264'),
```